Rating Prediction

Predicting AirBnB Review Scores

Ruth • Ido • Ella • Rosemary

The Challenge

Most ratings cluster at the top - very little variance to predict

Median: 4.82

Our Training Data

Two very different markets to ensure generalization

Los Angeles

47%
Superhost rate

1-6 days
Typical stay

Short vacation stays, tourists, weekend getaways

New York City

26%
Superhost rate

30+ days
Typical stay

Long-term rentals, relocations, work assignments

Model Evolution

Our path from overfitting to generalization

Baseline: 0.4058

XGBoost

0.44

Overfit badly

Embeddings

0.3818

Good but complex

Ridge 15

0.3816

Simpler is better

24 features with xgboost

0.378

key Insight: Simpler Ridge regression outperformed complex XGBoost when tested on unseen cities

Surprising Findings

What we expected to matter.. didn't

Price

Location

Feature Architecture

Simple, universal features that work across cities

PROPERTY

- accommodates
- bathrooms
- bedrooms
- beds
- room ratio

HOST

- response rate
- acceptance rate
- is superhost
- tenure
- response speed

TEXT

- has description
- desc length
- has host about

BOOKING

- minimum nights
- instant bookable

KEYWORDS

- clean
- luxury
- view
- location
- modern
- cozy
- quiet

Validation Performance

Evaluated on Chicago (unseen during training)

BASELINE

0.4058

RMSE

Predicts mean rating

OUR MODEL

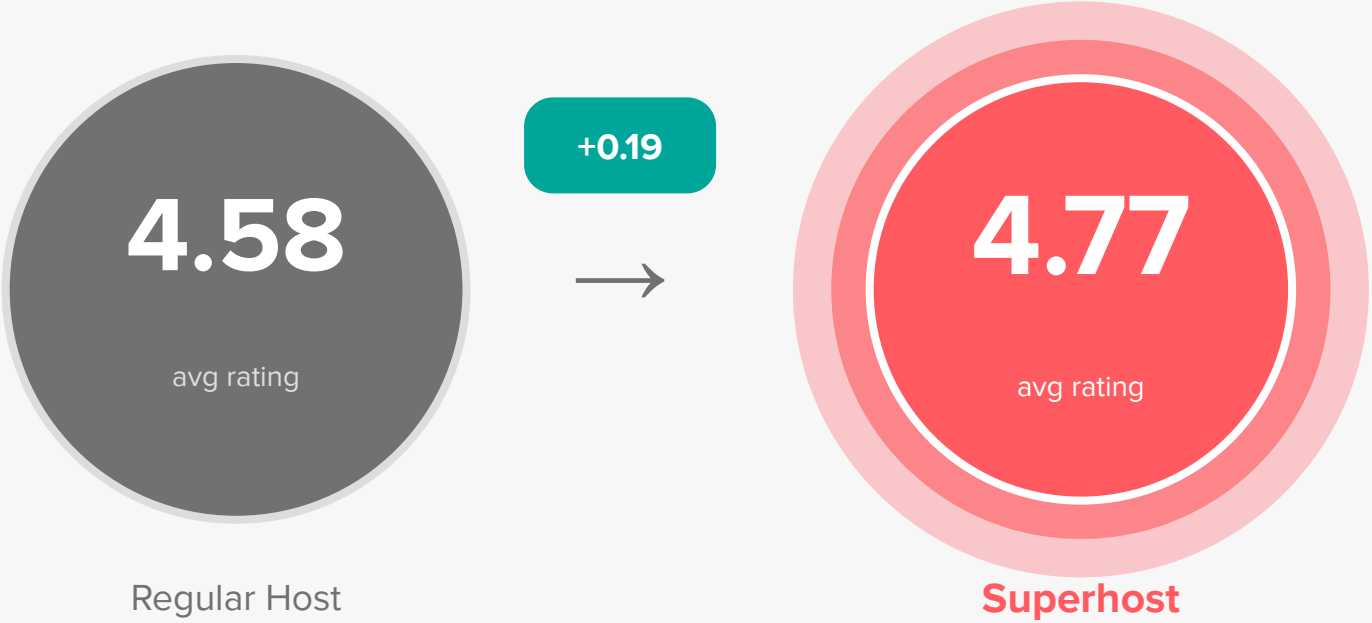
0.378

RMSE

-6.8%

Superhost Advantage

Consistent +0.19 rating boost across all cities



Test Set Performance

Tested on Sydney

BASELINE

0.4412

RMSE

Predicts mean rating

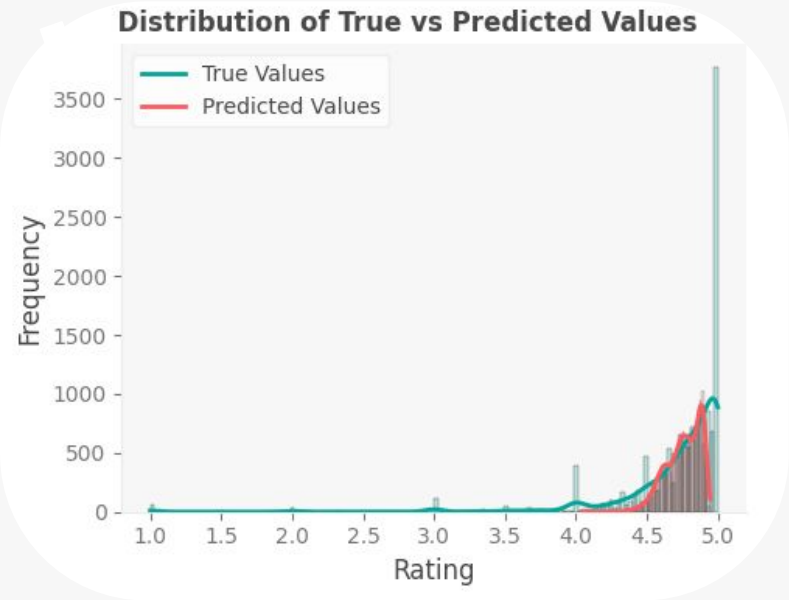
OUR MODEL

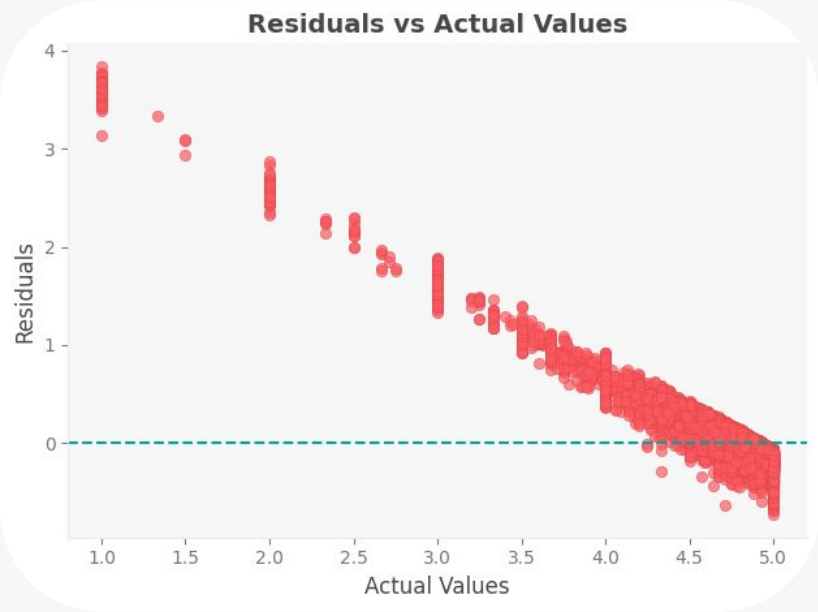
0.4122

RMSE

-6.6%

Error analysis





What We Learned

Key takeaways from our ML journey

1. Start Simple
2. Test Generalization
3. Visualize Features
4. Cache Everything
5. Balanced Data

Thank You

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How to Improve Your Ratings

Actionable tips based on our model findings

1

Respond Fast

Maintain 90%+ response rate. Reply to inquiries within hours, not days.

2

Never Cancel

Keep cancellation rate below 1%. Honor every booking commitment.

3

Polish Your Listing

Write detailed descriptions. Use keywords: clean, modern, cozy, view.

4

Aim for Superhost

The +0.19 rating boost compounds over time. It's worth the effort.

Focus on service, not spending — that's what moves ratings